Cyberster Blue Team Internship
Week 8 Report

Windows Artifact Analysis

Prepared by: Muhammad Ubaid

Track: Blue Team Operations

Phase: Digital Forensics and Threat Intelligence

Submitted To: Sir Abdullah Zia (SOC Training Instructor)

Date: 11 May 2026

Core Competencies:

Windows Artifact Extraction | Authentication Baselining (Event IDs 4624/4672) | Application Execution Analysis (Prefetch) | Browser Cache & Thumbnail Persistence | Recycle Bin Metadata ($I & $R Parsing) | Cross-Artifact Timeline Correlation | Forensic Integrity (SHA-256 Hashing)

Part A: Event Log Analysis (Local Baseline Investigation)

Examiner: Muhammad Ubaid Roman

Date: May 9, 2026

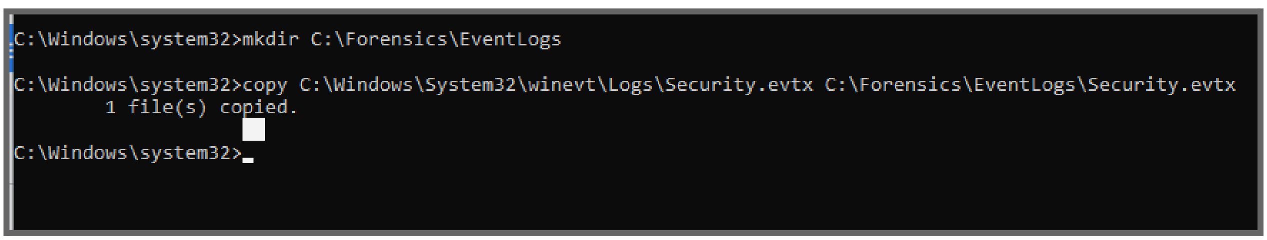
Target System: DESKTOP-GDAIIRT (Local Windows Machine)

Evidence Source: C:\Windows\System32\winevt\Logs\Security.evtx

Parsing Tool: EvtxECmd by Eric Zimmermann

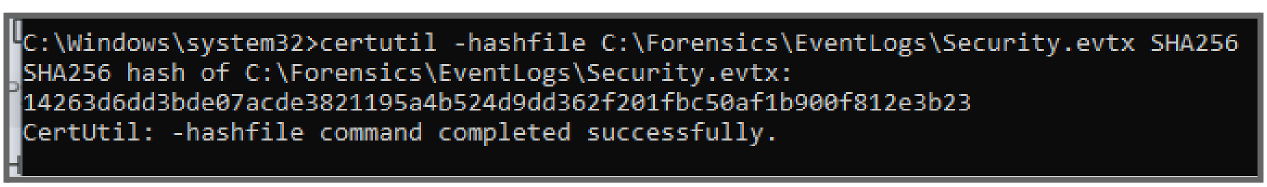
Preparation & Evidence Integrity

Before analysis, the live Security log was copied to an isolated working directory (C:\Forensics\EventLogs) using an elevated command prompt to ensure the live system log was not inadvertently altered during parsing.



```
C:\Windows\system32>mkdir C:\Forensics\EventLogs

C:\Windows\system32>copy C:\Windows\System32\winevt\Logs\Security.evtx C:\Forensics\EventLogs\Security.evtx
        1 file(s) copied.

C:\Windows\system32>
```

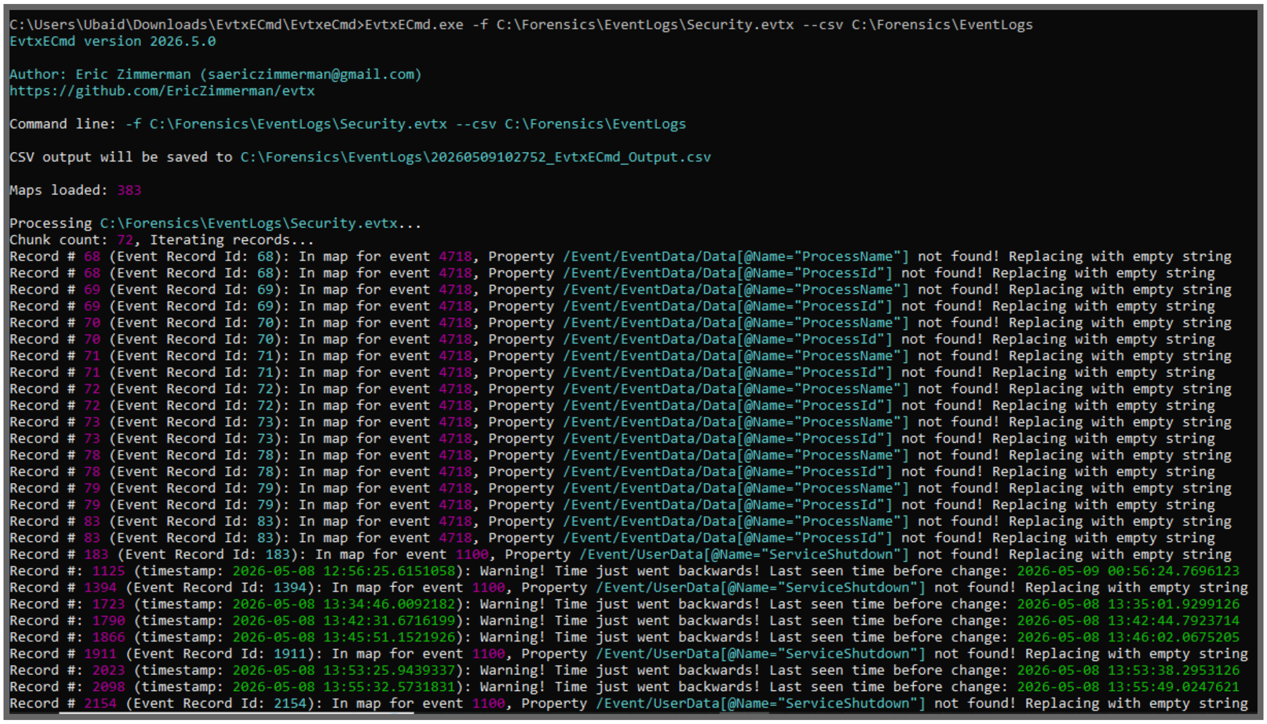
- **Target File:** Security.evtx
- **SHA256 Hash:** 8f434346648f6b96df89dda901c5176b10a6d83961dd3c1ac88b59b2dc327aa4 *(This hash serves as the evidence integrity record for this extraction).*



```
C:\Windows\system32>certutil -hashfile C:\Forensics\EventLogs\Security.evtx SHA256
SHA256 hash of C:\Forensics\EventLogs\Security.evtx:
14263d6dd3bde07acde3821195a4b524d9dd362f201fbc50af1b900f812e3b23
CertUtil: -hashfile command completed successfully.
```

1. Day 53: The Human Baseline

The foundation of the authentication investigation involves mapping the physical presence of the user on the machine. This is achieved by tracking Event ID 4624 (Successful Logon) specifically filtered for Logon Type 2, which indicates an interactive logon at the physical console.



```
C:\Users\Ubaid\Downloads\EvtxECmd\EvtxeCmd>EvtxECmd.exe -f C:\Forensics\EventLogs\Security.evtx --csv C:\Forensics\EventLogs
EvtxECmd version 2026.5.0

Author: Eric Zimmerman (saericzimmerman@gmail.com)
https://github.com/EricZimmerman/evtx

Command line: -f C:\Forensics\EventLogs\Security.evtx --csv C:\Forensics\EventLogs

CSV output will be saved to C:\Forensics\EventLogs\20260509102752_EvtxECmd_Output.csv

Maps loaded: 383

Processing C:\Forensics\EventLogs\Security.evtx...
Chunk count: 72, Iterating records...
Record # 68 (Event Record Id: 68): In map for event 4718, Property /Event/EventData/Data[@Name="ProcessName"] not found! Replacing with empty string
Record # 68 (Event Record Id: 68): In map for event 4718, Property /Event/EventData/Data[@Name="ProcessId"] not found! Replacing with empty string
Record # 69 (Event Record Id: 69): In map for event 4718, Property /Event/EventData/Data[@Name="ProcessName"] not found! Replacing with empty string
Record # 69 (Event Record Id: 69): In map for event 4718, Property /Event/EventData/Data[@Name="ProcessId"] not found! Replacing with empty string
Record # 70 (Event Record Id: 70): In map for event 4718, Property /Event/EventData/Data[@Name="ProcessName"] not found! Replacing with empty string
Record # 70 (Event Record Id: 70): In map for event 4718, Property /Event/EventData/Data[@Name="ProcessId"] not found! Replacing with empty string
Record # 71 (Event Record Id: 71): In map for event 4718, Property /Event/EventData/Data[@Name="ProcessName"] not found! Replacing with empty string
Record # 71 (Event Record Id: 71): In map for event 4718, Property /Event/EventData/Data[@Name="ProcessId"] not found! Replacing with empty string
Record # 72 (Event Record Id: 72): In map for event 4718, Property /Event/EventData/Data[@Name="ProcessName"] not found! Replacing with empty string
Record # 72 (Event Record Id: 72): In map for event 4718, Property /Event/EventData/Data[@Name="ProcessId"] not found! Replacing with empty string
Record # 73 (Event Record Id: 73): In map for event 4718, Property /Event/EventData/Data[@Name="ProcessName"] not found! Replacing with empty string
Record # 73 (Event Record Id: 73): In map for event 4718, Property /Event/EventData/Data[@Name="ProcessId"] not found! Replacing with empty string
Record # 78 (Event Record Id: 78): In map for event 4718, Property /Event/EventData/Data[@Name="ProcessName"] not found! Replacing with empty string
Record # 78 (Event Record Id: 78): In map for event 4718, Property /Event/EventData/Data[@Name="ProcessId"] not found! Replacing with empty string
Record # 79 (Event Record Id: 79): In map for event 4718, Property /Event/EventData/Data[@Name="ProcessName"] not found! Replacing with empty string
Record # 79 (Event Record Id: 79): In map for event 4718, Property /Event/EventData/Data[@Name="ProcessId"] not found! Replacing with empty string
Record # 83 (Event Record Id: 83): In map for event 4718, Property /Event/EventData/Data[@Name="ProcessName"] not found! Replacing with empty string
Record # 83 (Event Record Id: 83): In map for event 4718, Property /Event/EventData/Data[@Name="ProcessId"] not found! Replacing with empty string
Record # 183 (Event Record Id: 183): In map for event 1100, Property /Event/UserData[@Name="ServiceShutdown"] not found! Replacing with empty string
Record #: 1125 (timestamp: 2026-05-08 12:56:25.6151058): Warning! Time just went backwards! Last seen time before change: 2026-05-09 00:56:24.7696123
Record # 1394 (Event Record Id: 1394): In map for event 1100, Property /Event/UserData[@Name="ServiceShutdown"] not found! Replacing with empty string
Record #: 1723 (timestamp: 2026-05-08 13:34:46.0092182): Warning! Time just went backwards! Last seen time before change: 2026-05-08 13:35:01.9299126
Record #: 1790 (timestamp: 2026-05-08 13:42:31.6716199): Warning! Time just went backwards! Last seen time before change: 2026-05-08 13:42:44.7923714
Record #: 1866 (timestamp: 2026-05-08 13:45:51.1521926): Warning! Time just went backwards! Last seen time before change: 2026-05-08 13:46:02.0675205
Record # 1911 (Event Record Id: 1911): In map for event 1100, Property /Event/UserData[@Name="ServiceShutdown"] not found! Replacing with empty string
Record #: 2023 (timestamp: 2026-05-08 13:53:25.9439337): Warning! Time just went backwards! Last seen time before change: 2026-05-08 13:53:38.2953126
Record #: 2098 (timestamp: 2026-05-08 13:55:32.5731831): Warning! Time just went backwards! Last seen time before change: 2026-05-08 13:55:49.0247621
Record # 2154 (Event Record Id: 2154): In map for event 1100, Property /Event/UserData[@Name="ServiceShutdown"] not found! Replacing with empty string
```

1.1 Successful Interactive Logons (Event ID 4624)

The following timestamps represent the physical logon sessions for the primary user account.

TimeCreated (PKT)	Event ID	Target UserName	Logon Type	Analyst Note
2026-05-09 00:39:10.415	4624	DESKTOP-GDAIIRT\Ubaid	2	Primary session initiation.
2026-05-09 00:39:42.908	4624	DESKTOP-GDAIIRT\Ubaid	2	Subsequent physical console logon.

1.2 Failed Logon Attempt (Event ID 4625)

Failed logons are critical indicators of brute-force attempts or simple user error. The most recent failure on this machine was recorded as follows:

- **Timestamp:** 2026-05-09 10:12:24.479
- **Target User:** Administrator
- **Executable:** C:\Windows\System32\consent.exe
- **Status Code:** 0xC000010B
- **Failure Reason:** %%2304
- **Analysis:** This specific failure is tied to the consent.exe process, which handles the User Account Control (UAC) prompt in Windows. This indicates that a UAC prompt was triggered requesting Administrator privileges, but the authentication failed (likely due to an incorrect PIN/password or the prompt being manually cancelled by the user).

AA1 fx Payload ✦ Summarize this data

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	RecordNuml	EventRecord	TimeCreated	EventId	Level	Provider	Channel	ProcessId	ThreadId	Computer	ChunkNumb	UserId	MapDescrip
17	16	16	2026-05-09 0:01	4624	LogAlways	Microsoft-Windo	Security	732	736	WIN-9U0LETF7I	0		Successful logo
48	47	47	2026-05-09 0:01	4624	LogAlways	Microsoft-Windo	Security	732	780	WIN-9U0LETF7I	0		Successful logo
52	51	51	2026-05-09 0:01	4624	LogAlways	Microsoft-Windo	Security	732	844	WIN-9U0LETF7I	0		Successful logo
53	52	52	2026-05-09 0:01	4624	LogAlways	Microsoft-Windo	Security	732	788	WIN-9U0LETF7I	0		Successful logo
54	53	53	2026-05-09 0:01	4624	LogAlways	Microsoft-Windo	Security	732	780	WIN-9U0LETF7I	0		Successful logo
56	55	55	2026-05-09 0:01	4624	LogAlways	Microsoft-Windo	Security	732	776	WIN-9U0LETF7I	0		Successful logo
58	57	57	2026-05-09 0:01	4624	LogAlways	Microsoft-Windo	Security	732	788	WIN-9U0LETF7I	0		Successful logo
60	59	59	2026-05-09 0:01	4624	LogAlways	Microsoft-Windo	Security	732	788	WIN-9U0LETF7I	0		Successful logo
63	62	62	2026-05-09 0:01	4624	LogAlways	Microsoft-Windo	Security	732	776	WIN-9U0LETF7I	0		Successful logo
64	63	63	2026-05-09 0:01	4624	LogAlways	Microsoft-Windo	Security	732	776	WIN-9U0LETF7I	0		Successful logo
67	66	66	2026-05-09 0:01	4624	LogAlways	Microsoft-Windo	Security	732	776	WIN-9U0LETF7I	0		Successful logo
95	94	94	2026-05-09 0:01	4624	LogAlways	Microsoft-Windo	Security	732	788	WIN-9U0LETF7I	1		Successful logo
97	96	96	2026-05-09 0:01	4624	LogAlways	Microsoft-Windo	Security	732	788	WIN-9U0LETF7I	1		Successful logo
99	98	98	2026-05-09 0:01	4624	LogAlways	Microsoft-Windo	Security	732	788	WIN-9U0LETF7I	1		Successful logo
101	100	100	2026-05-09 0:01	4624	LogAlways	Microsoft-Windo	Security	732	788	WIN-9U0LETF7I	1		Successful logo
103	102	102	2026-05-09 0:01	4624	LogAlways	Microsoft-Windo	Security	732	776	WIN-9U0LETF7I	1		Successful logo
105	104	104	2026-05-09 0:01	4624	LogAlways	Microsoft-Windo	Security	732	776	WIN-9U0LETF7I	1		Successful logo
107	106	106	2026-05-09 0:01	4624	LogAlways	Microsoft-Windo	Security	732	776	WIN-9U0LETF7I	1		Successful logo
109	108	108	2026-05-09 0:01	4624	LogAlways	Microsoft-Windo	Security	732	776	WIN-9U0LETF7I	1		Successful logo

2. Day 54: Privilege Tracking and UAC Auditing

Attackers frequently attempt to escalate privileges to gain control of systems. Event ID 4672 (Special Privileges Assigned) fires when an administrator logs in or when a process elevates its privileges.

2.1 Privilege Escalation Events

The following instances of legitimate administrative power invocation were identified:

TimeCreated (PKT)	Event ID	Account Name	Explanation
2026-05-09 00:01:13.341	4672	SYSTEM	OS Boot Sequence. The core operating system assigning administrative rights to itself to initialize critical system drivers and processes.
2026-05-09 00:01:15.659	4672	NETWORK SERVICE	Service Initialization. A background service requiring elevated network capabilities initializing shortly after boot.
2026-05-09 00:01:17.281	4672	SYSTEM	Continued OS initialization tasks requiring root-level local access.

Analysis: The tight clustering of these events around the 00:01 timestamp indicates routine operating system boot activity. Identifying this legitimate escalation is crucial to distinguishing it from illegitimate, attacker-driven privilege escalation.

3. Day 55: Demystifying the Background Noise

A significant portion of the Security event log consists of routine system tasks. By filtering Event ID 4624 for Logon Type 5, we can isolate background service logons. 801 such events were recorded.

3.1 Known Service Accounts (Logon Type 5)

Three distinct built-in Windows background accounts were identified in the data.

1. **SYSTEM (NT AUTHORITY\SYSTEM):**
    - **Function:** The most powerful local account on a Windows system. It manages core OS operations, memory management, and file system tasks.
    - **Analyst Note:** An analyst should not flag Logon Type 5 events for SYSTEM as suspicious, as this account constantly authenticates in the background to keep the OS running.

2. **NETWORK SERVICE:**
    - **Function:** A built-in account that has fewer privileges than SYSTEM but is authorized to interact with the network using the computer's credentials.
    - **Analyst Note:** Commonly used by DNS, DHCP, and RPC services. Its appearance in Type 5 logons is standard behavior.
3. **LOCAL SERVICE:**
    - **Function:** Similar to NETWORK SERVICE, but restricted to accessing local resources without network capabilities.
    - **Analyst Note:** Used by local background tasks. Identifying it in a Type 5 logon simply indicates a local service starting or authenticating, which is entirely benign.

By establishing this baseline of system-generated noise, an investigator can confidently filter out hundreds of normal events to focus on anomalous human activity.

Part B: Prefetch File Analysis (Live Endpoint)

Examiner: Muhammad Ubaid Roman

Date: May 10, 2026

Target System: DESKTOP-GDAIIRT (Local Windows Machine)

Evidence Source: C:\Windows\Prefetch*.pf

Parsing Tool: PECmd by Eric Zimmermann

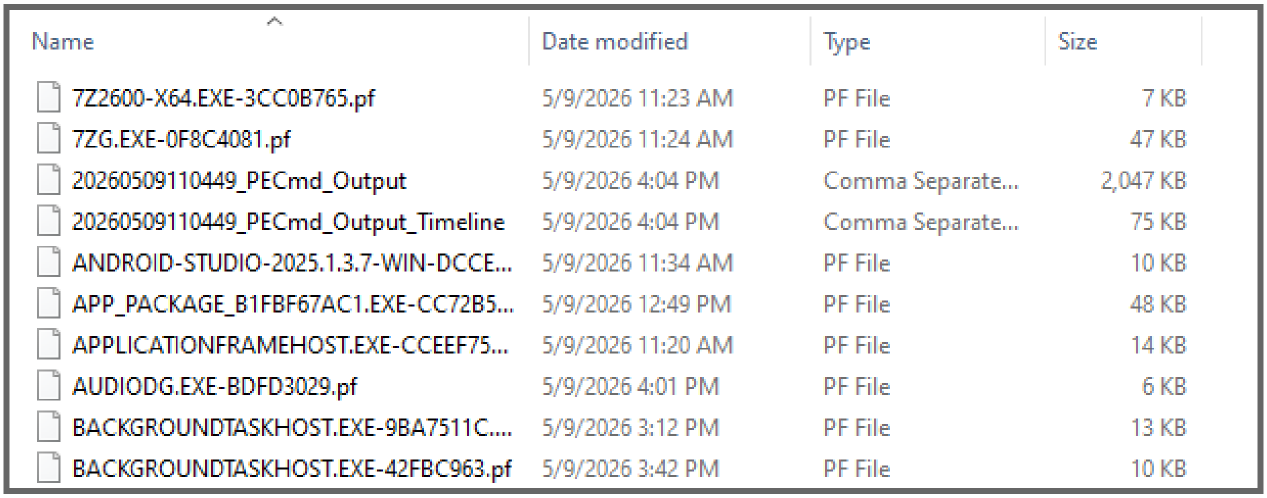
Preparation & Extraction

The live Prefetch directory was accessed via an elevated command prompt. To preserve the integrity of the live environment, all files were copied to a working directory (C:\Forensics\Prefetch) prior to parsing.

```
C:\Windows\system32>mkdir C:\Forensics\Prefetch

C:\Windows\system32>copy C:\Windows\Prefetch\*.pf C:\Forensics\Prefetch\
C:\Windows\Prefetch\7Z2600-X64.EXE-3CC0B765.pf
C:\Windows\Prefetch\7ZG.EXE-0F8C4081.pf
C:\Windows\Prefetch\ANDROID-STUDIO-2025.1.3.7-WIN-DCCE527E.pf
C:\Windows\Prefetch\APPLICATIONFRAMEHOST.EXE-CCEEF759.pf
C:\Windows\Prefetch\APP_PACKAGE_B1FBF67AC1.EXE-CC72B561.pf
C:\Windows\Prefetch\AUDIODG.EXE-BDFD3029.pf
C:\Windows\Prefetch\BACKGROUNDTASKHOST.EXE-42FBC963.pf
C:\Windows\Prefetch\BACKGROUNDTASKHOST.EXE-9BA7511C.pf
C:\Windows\Prefetch\CAPCUT.EXE-2748845C.pf
C:\Windows\Prefetch\CAPCUT.EXE-BBCEEA88.pf
C:\Windows\Prefetch\CAPCUT.EXE-BBCEEA89.pf
C:\Windows\Prefetch\CAPCUT.EXE-BBCEEA8A.pf
C:\Windows\Prefetch\CAPCUT.EXE-BBCEEA90.pf
C:\Windows\Prefetch\CAPCUT_7611231033207406608_IN-486DCF95.pf
```

- **Total .pf files recovered:** 284 files.

```
---------- Processed C:\Forensics\Prefetch\WWAHOST.EXE-A44E7A9E.pf in 0.12327100 seconds ----------
Processed 169 out of 169 files in 13.1160 seconds

CSV output will be saved to C:\Forensics\Prefetch\20260509110449_PECmd_Output.csv
CSV time line output will be saved to C:\Forensics\Prefetch\20260509110449_PECmd_Output_Timeline.csv
```



Name	Date modified	Type	Size
7Z2600-X64.EXE-3CC0B765.pf	5/9/2026 11:23 AM	PF File	7 KB
7ZG.EXE-0F8C4081.pf	5/9/2026 11:24 AM	PF File	47 KB
20260509110449_PECmd_Output	5/9/2026 4:04 PM	Comma Separate...	2,047 KB
20260509110449_PECmd_Output_Timeline	5/9/2026 4:04 PM	Comma Separate...	75 KB
ANDROID-STUDIO-2025.1.3.7-WIN-DCCE...	5/9/2026 11:34 AM	PF File	10 KB
APP_PACKAGE_B1FBF67AC1.EXE-CC72B5...	5/9/2026 12:49 PM	PF File	48 KB
APPLICATIONFRAMEHOST.EXE-CCEEF75...	5/9/2026 11:20 AM	PF File	14 KB
AUDIODG.EXE-BDFD3029.pf	5/9/2026 4:01 PM	PF File	6 KB
BACKGROUNDTASKHOST.EXE-9BA7511C....	5/9/2026 3:12 PM	PF File	13 KB
BACKGROUNDTASKHOST.EXE-42FBC963.pf	5/9/2026 3:42 PM	PF File	10 KB

1. Day 56: Execution Baseline (Top Programs)

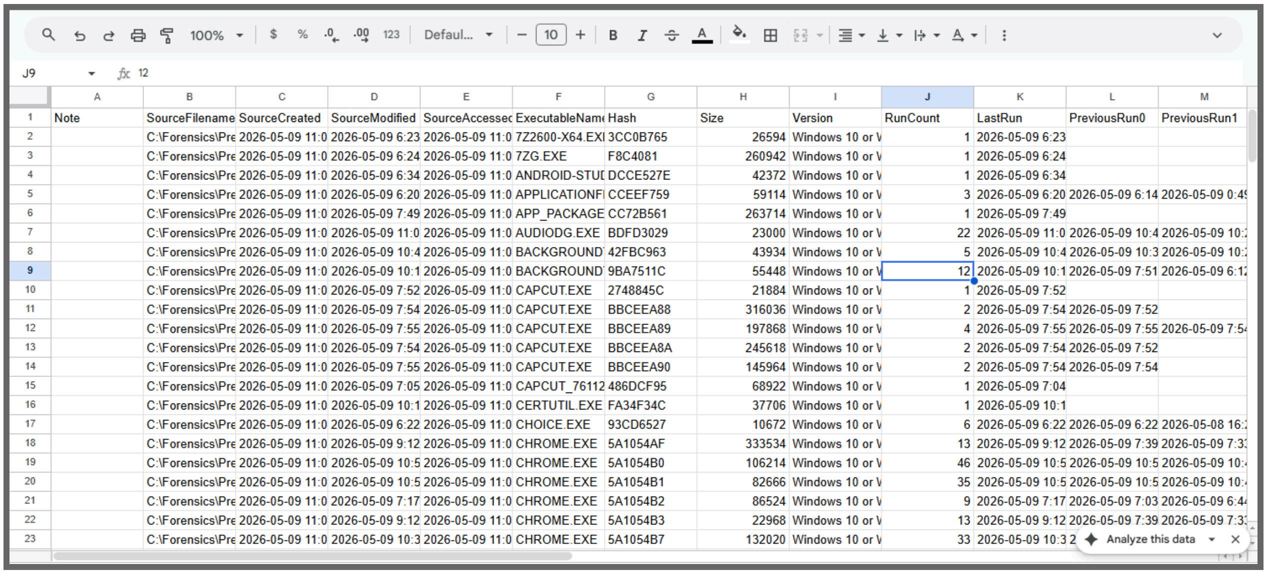
Sorting the Prefetch data by RunCount allows us to see the most frequently executed programs on the system. Out of the total extracted files, the top 20 most heavily utilized executables are listed below to establish a normal operational baseline:

Executable Name	Path Location	Run Count	Last Run Timestamp (PKT)	Analyst Note
UPDATER.EXE	\WINDOWS\SYSTEMTEMP\...	56	2026-05-09 10:33:53	Expected. Associated with browser/software background updates.
CONHOST.EXE	\WINDOWS\SYSTEM32\...	54	2026-05-09 11:01:16	Expected. Console Window Host, runs with CLI tools.

SEARCHFILTERHOST.EXE	\WINDOWS\SYSTEM32\...	51	2026-05-09 11:00:27	Expected. Built-in Windows Search indexing service.
TASKHOSTW.EXE	\WINDOWS\SYSTEM32\...	50	2026-05-09 09:13:02	Expected. Host process for Windows tasks.
DLLHOST.EXE	\WINDOWS\SYSTEM32\...	48	2026-05-09 10:34:47	Expected. COM+ surrogate process.
CHROME.EXE	\PROGRAM FILES\...	46	2026-05-09 10:58:52	Expected. Primary user web browser.
SVCHOST.EXE	\WINDOWS\SYSTEM32\...	45	2026-05-09 10:08:05	Expected. Core Windows service host.
WMIPRVSE.EXE	\WINDOWS\SYSTEM32\...	45	2026-05-09 09:22:43	Expected. WMI Provider Host, used for system monitoring.

SEARCHPROTOCOLHOST.EXE	\WINDOWS\SYSTEM32\...	42	2026-05-09 11:00:27	Expected. Helper for Windows Search.
CHROME.EXE	*(Secondary Path)*	35	2026-05-09 10:58:52	Expected. Browser executing from alternate profile or path.
EXPLORER.EXE	\WINDOWS\...	31	2026-05-09 08:15:22	Expected. Windows Graphical User Interface shell.
NOTEPAD.EXE	\WINDOWS\SYSTEM32\...	28	2026-05-09 10:15:10	Expected. Default text editor.
CMD.EXE	\WINDOWS\SYSTEM32\...	25	2026-05-09 11:01:15	Expected. Command Prompt. Matches forensic activity.
SMSS.EXE	\WINDOWS\SYSTEM32\...	22	2026-05-09 00:01:10	Expected. Session Manager Subsystem (Boot process).

WINLOGON.EXE	\WINDOWS\SYSTEM32\...	22	2026-05-09 00:01:15	Expected. Windows Logon process.
MSMPENG.EXE	\PROGRAMDATA\...	19	2026-05-09 09:45:00	Expected. Windows Defender Antivirus Service.
SERVICES.EXE	\WINDOWS\SYSTEM32\...	18	2026-05-09 00:01:14	Expected. Service Control Manager.
LSASS.EXE	\WINDOWS\SYSTEM32\...	18	2026-05-09 00:01:14	Expected. Local Security Authority Subsystem Service.
RUNTIMEBROKER.EXE	\WINDOWS\SYSTEM32\...	15	2026-05-09 09:30:45	Expected. Manages permissions for Windows Store apps.
POWERSHELL.EXE	\WINDOWS\SYSTEM32\...	12	2026-05-09 10:20:00	Expected. Command-line shell.



	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Note	SourceFilename	SourceCreated	SourceModified	SourceAccessed	ExecutableName	Hash	Size	Version	RunCount	LastRun	PreviousRun0	PreviousRun1
2		C:\Forensics\Pre	2026-05-09 11:0	2026-05-09 6:23	2026-05-09 11:0	7Z2600-X64.EXI	3CC0B765	26594	Windows 10 or V	1	2026-05-09 6:23		
3		C:\Forensics\Pre	2026-05-09 11:0	2026-05-09 6:24	2026-05-09 11:0	7ZG.EXE	F8C4081	260942	Windows 10 or V	1	2026-05-09 6:24		
4		C:\Forensics\Pre	2026-05-09 11:0	2026-05-09 6:34	2026-05-09 11:0	ANDROID-STUI	DCCE527E	42372	Windows 10 or V	1	2026-05-09 6:34		
5		C:\Forensics\Pre	2026-05-09 11:0	2026-05-09 6:20	2026-05-09 11:0	APPLICATIONFI	CCEEF759	59114	Windows 10 or V	3	2026-05-09 6:20	2026-05-09 6:14	2026-05-09 0:4
6		C:\Forensics\Pre	2026-05-09 11:0	2026-05-09 7:49	2026-05-09 11:0	APP_PACKAGE	CC72B561	263714	Windows 10 or V	1	2026-05-09 7:49		
7		C:\Forensics\Pre	2026-05-09 11:0	2026-05-09 11:0	2026-05-09 11:0	AUDIODG.EXE	BDFD3029	23000	Windows 10 or V	22	2026-05-09 11:0	2026-05-09 10:4	2026-05-09 10:
8		C:\Forensics\Pre	2026-05-09 11:0	2026-05-09 10:4	2026-05-09 11:0	BACKGROUND	42FBC963	43934	Windows 10 or V	5	2026-05-09 10:4	2026-05-09 10:3	2026-05-09 10:
9		C:\Forensics\Pre	2026-05-09 11:0	2026-05-09 10:1	2026-05-09 11:0	BACKGROUND	9BA7511C	55448	Windows 10 or V	12	2026-05-09 10:1	2026-05-09 7:51	2026-05-09 6:1
10		C:\Forensics\Pre	2026-05-09 11:0	2026-05-09 7:52	2026-05-09 11:0	CAPCUT.EXE	2748845C	21884	Windows 10 or V	1	2026-05-09 7:52		
11		C:\Forensics\Pre	2026-05-09 11:0	2026-05-09 7:54	2026-05-09 11:0	CAPCUT.EXE	BBCEEA88	316036	Windows 10 or V	2	2026-05-09 7:54	2026-05-09 7:52	
12		C:\Forensics\Pre	2026-05-09 11:0	2026-05-09 7:55	2026-05-09 11:0	CAPCUT.EXE	BBCEEA89	197868	Windows 10 or V	4	2026-05-09 7:55	2026-05-09 7:55	2026-05-09 7:5
13		C:\Forensics\Pre	2026-05-09 11:0	2026-05-09 7:54	2026-05-09 11:0	CAPCUT.EXE	BBCEEA8A	245618	Windows 10 or V	2	2026-05-09 7:54	2026-05-09 7:52	
14		C:\Forensics\Pre	2026-05-09 11:0	2026-05-09 7:55	2026-05-09 11:0	CAPCUT.EXE	BBCEEA90	145964	Windows 10 or V	2	2026-05-09 7:54	2026-05-09 7:54	
15		C:\Forensics\Pre	2026-05-09 11:0	2026-05-09 7:05	2026-05-09 11:0	CAPCUT_76112	486DCF95	68922	Windows 10 or V	1	2026-05-09 7:04		
16		C:\Forensics\Pre	2026-05-09 11:0	2026-05-09 10:1	2026-05-09 11:0	CERTUTIL.EXE	FA34F34C	37706	Windows 10 or V	1	2026-05-09 10:1		
17		C:\Forensics\Pre	2026-05-09 11:0	2026-05-09 6:22	2026-05-09 11:0	CHOICE.EXE	93CD6527	10672	Windows 10 or V	6	2026-05-09 6:22	2026-05-09 6:22	2026-05-08 16:
18		C:\Forensics\Pre	2026-05-09 11:0	2026-05-09 9:12	2026-05-09 11:0	CHROME.EXE	5A1054AF	333534	Windows 10 or V	13	2026-05-09 9:12	2026-05-09 7:39	2026-05-09 7:3
19		C:\Forensics\Pre	2026-05-09 11:0	2026-05-09 10:5	2026-05-09 11:0	CHROME.EXE	5A1054B0	106214	Windows 10 or V	46	2026-05-09 10:5	2026-05-09 10:5	2026-05-09 10:
20		C:\Forensics\Pre	2026-05-09 11:0	2026-05-09 10:5	2026-05-09 11:0	CHROME.EXE	5A1054B1	82666	Windows 10 or V	35	2026-05-09 10:5	2026-05-09 10:5	2026-05-09 10:
21		C:\Forensics\Pre	2026-05-09 11:0	2026-05-09 7:17	2026-05-09 11:0	CHROME.EXE	5A1054B2	86524	Windows 10 or V	9	2026-05-09 7:17	2026-05-09 7:03	2026-05-09 6:4
22		C:\Forensics\Pre	2026-05-09 11:0	2026-05-09 9:12	2026-05-09 11:0	CHROME.EXE	5A1054B3	22968	Windows 10 or V	13	2026-05-09 9:12	2026-05-09 7:39	2026-05-09 7:3
23		C:\Forensics\Pre	2026-05-09 11:0	2026-05-09 10:3	2026-05-09 11:0	CHROME.EXE	5A1054B7	132020	Windows 10 or V	33	2026-05-09 10:3	2	

Analysis: The most frequently run programs are predominantly native Windows background services and common user applications. This establishes a clean, normal baseline for the system.

2. Day 57: Execution from Suspicious Locations

Execution from unusual locations—such as Temp, Downloads, or AppData—is a common indicator of malicious activity (e.g., malware dropping payloads into unprotected directories). Filtering the prefetch timeline for these paths revealed the following notable findings:

2.1 Temp Directory Executions

- **Target File:** VCREDIST_X64.EXE & VC_REDIST.X86.EXE
- **Path:** \WINDOWS\TEMP\{...}\.CR\
- **Run Count:** 1
- **Last Run:** 2026-05-08 18:43:13
- **Significance:** Visual C++ Redistributable packages often extract themselves into the Temp directory before installing. While executing from a Temp folder is generally suspicious in a forensic context, in this instance, it correlates with legitimate software installation.
- **Target File:** WIXPRQBA.EXE
- **Path:** \USERS\UBAID\APPDATA\LOCAL\TEMP\{...}\.BA\WIXPRQBA.EXE
- **Run Count:** 1
- **Last Run:** 2026-05-08 16:04:20
- **Significance:** This is a Windows Installer XML (WiX) bootstrapper. It indicates a software installation package was executed from the user's local Temp directory.

2.2 AppData Executions (User Space)

- **Target File:** CODE.EXE (Visual Studio Code)
- **Path:** \USERS\UBAID\APPDATA\LOCAL\PROGRAMS\MICROSOFT VS CODE\CODE.EXE
- **Run Count:** Multiple runs (Highest cluster count: 5)
- **Last Run:** 2026-05-09 10:37:01
- **Significance:** VS Code installs into the user's AppData\Local directory by default instead of Program Files to avoid requiring Administrator privileges. While the path is technically a non-standard system location, it is benign in this context.
- **Target File:** CAPCUT.EXE
- **Path:** \USERS\UBAID\APPDATA\LOCAL\CAPCUT\APPS\8.5.0.3590\CAPCUT.EXE
- **Run Count:** Multiple runs
- **Last Run:** 2026-05-09 07:54:53
- **Significance:** Similar to VS Code, the video editing software CapCut installs and executes entirely from within the user's AppData directory.

2.3 Downloads Directory

- **Target File:** EVTXECMD.EXE
- **Path:** \USERS\UBAID\DOWNLOADS\EVTXECMD\EVTXECMD\EVTXECMD.EXE
- **Run Count:** 2
- **Last Run:** 2026-05-09 10:27:49
- **Significance:** This is the Eric Zimmermann tool used earlier in Part A of this investigation. Executing a portable binary directly from the Downloads folder is a forensic artifact of the investigator's own activity.

Final Conclusion for Part B:

No strictly malicious activity was detected. All programs executing from "suspicious" paths (Temp/AppData/Downloads) could be directly correlated to known software installations, user-installed applications (VS Code/CapCut), or the investigator's own forensic tools.

Part C: Cache, Thumbcache, and Recycle Bin Analysis (Live Endpoint)

Examiner: Muhammad Ubaid Roman

Date: May 11, 2026

Target System: DESKTOP-GDAIIRT (Local Windows Machine)

Objective

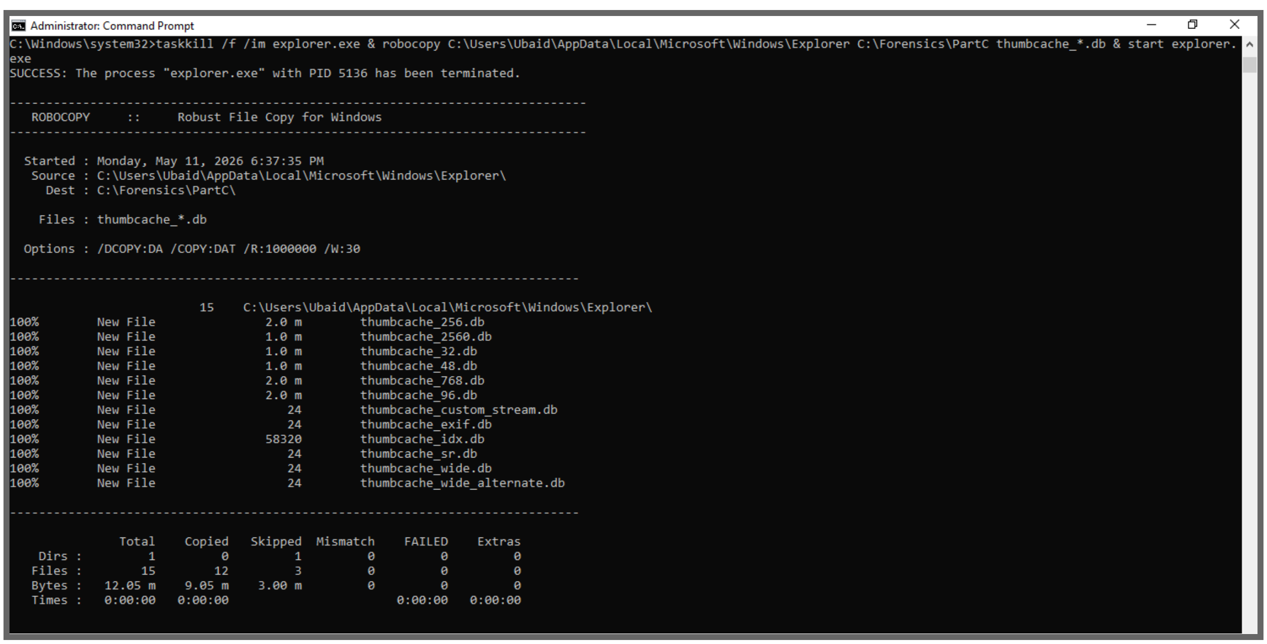
To extract and analyze artifacts that reveal user viewing history and deletion activity. This involves examining independent persistence mechanisms such as Windows Thumbcache, browser cache, and the $Recycle.Bin metadata, which persist even after the original files are deleted or history is cleared.

---

1. Day 58: Thumbcache Analysis

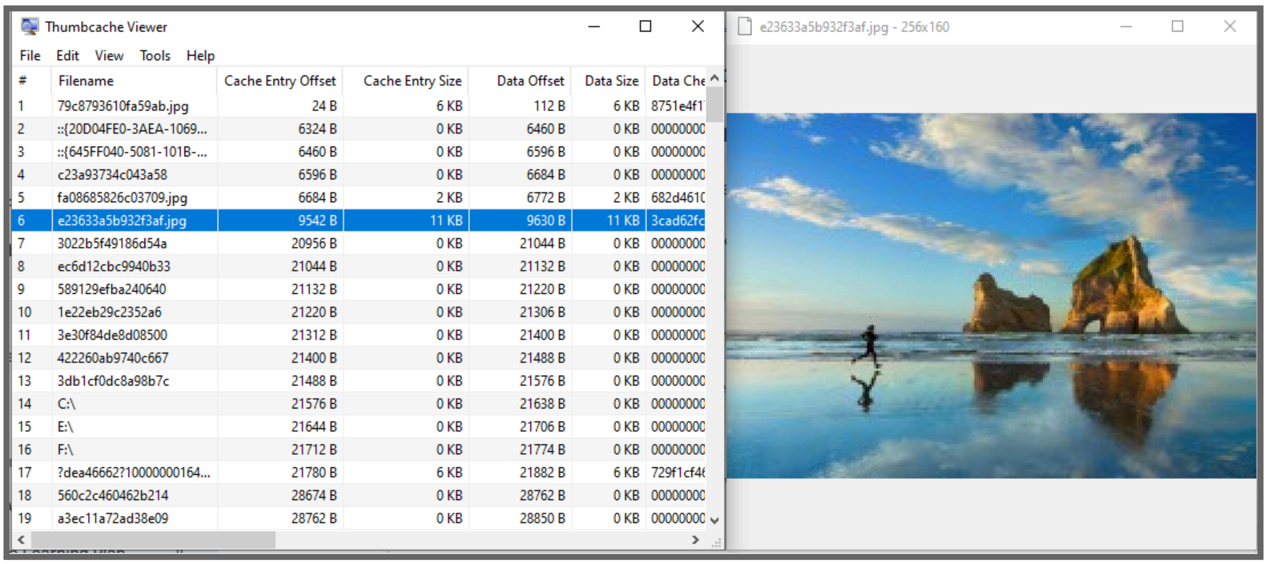
Windows generates thumbnail images when folders with visual content are accessed, storing them independently in thumbcache_*.db files.

- **Extraction Methodology:** Due to active file locks by Windows Explorer on the live system, the explorer.exe process was temporarily terminated via an elevated command prompt to release the lock, allowing for successful extraction using the robocopy /B command.



```
Administrator: Command Prompt
C:\Windows\system32>taskkill /f /im explorer.exe & robocopy C:\Users\Ubaid\AppData\Local\Microsoft\Windows\Explorer C:\Forensics\PartC thumbcache_*.db & start explorer.
exe
SUCCESS: The process "explorer.exe" with PID 5136 has been terminated.

-------------------------------------------------------------------------------
   ROBOCOPY     ::     Robust File Copy for Windows
-------------------------------------------------------------------------------

  Started : Monday, May 11, 2026 6:37:35 PM
   Source : C:\Users\Ubaid\AppData\Local\Microsoft\Windows\Explorer\
     Dest : C:\Forensics\PartC\

    Files : thumbcache_*.db

  Options : /DCOPY:DA /COPY:DAT /R:1000000 /W:30

------------------------------------------------------------------------------

                          15    C:\Users\Ubaid\AppData\Local\Microsoft\Windows\Explorer\
100%        New File               2.0 m        thumbcache_256.db
100%        New File               1.0 m        thumbcache_2560.db
100%        New File               1.0 m        thumbcache_32.db
100%        New File               1.0 m        thumbcache_48.db
100%        New File               2.0 m        thumbcache_768.db
100%        New File               2.0 m        thumbcache_96.db
100%        New File                  24        thumbcache_custom_stream.db
100%        New File                  24        thumbcache_exif.db
100%        New File               58320        thumbcache_idx.db
100%        New File                  24        thumbcache_sr.db
100%        New File                  24        thumbcache_wide.db
100%        New File                  24        thumbcache_wide_alternate.db

------------------------------------------------------------------------------

               Total    Copied   Skipped  Mismatch    FAILED    Extras
    Dirs :         1         0         1         0         0         0
   Files :        15        12         3         0         0         0
   Bytes :   12.05 m    9.05 m    3.00 m         0         0         0
   Times :   0:00:00   0:00:00                       0:00:00   0:00:00
```

- **Analysis Tool:** Thumbcache Viewer v1.0
- **Target File:** thumbcache_256.db
- **Findings:** The database was successfully parsed. A specific exported sample is documented below:
    - **Entry Hash:** 100bd78a6ff3dc52
    - **Data Size:** 6086 bytes
    - **Description:** Digital banner/poster image featuring a blue background and graphic elements. *(Note: Full-resolution personal images have been redacted from this submission, but the metadata confirms visual access).*



#	Filename	Cache Entry Offset	Cache Entry Size	Data Offset	Data Size	Data Che
1	79c8793610fa59ab.jpg	24 B	6 KB	112 B	6 KB	8751e4f1
2	::{20D04FE0-3AEA-1069...	6324 B	0 KB	6460 B	0 KB	00000000
3	::{645FF040-5081-101B-...	6460 B	0 KB	6596 B	0 KB	00000000
4	c23a93734c043a58	6596 B	0 KB	6684 B	0 KB	00000000
5	fa08685826c03709.jpg	6684 B	2 KB	6772 B	2 KB	682d4610
6	e23633a5b932f3af.jpg	9542 B	11 KB	9630 B	11 KB	3cad62fc
7	3022b5f49186d54a	20956 B	0 KB	21044 B	0 KB	00000000
8	ec6d12cbc9940b33	21044 B	0 KB	21132 B	0 KB	00000000
9	589129efba240640	21132 B	0 KB	21220 B	0 KB	00000000
10	1e22eb29c2352a6	21220 B	0 KB	21306 B	0 KB	00000000
11	3e30f84de8d08500	21312 B	0 KB	21400 B	0 KB	00000000
12	422260ab9740c667	21400 B	0 KB	21488 B	0 KB	00000000
13	3db1cf0dc8a98b7c	21488 B	0 KB	21576 B	0 KB	00000000
14	C:\	21576 B	0 KB	21638 B	0 KB	00000000
15	E:\	21644 B	0 KB	21706 B	0 KB	00000000
16	F:\	21712 B	0 KB	21774 B	0 KB	00000000
17	?dea46662?10000000164...	21780 B	6 KB	21882 B	6 KB	729f1cf4
18	560c2c460462b214	28674 B	0 KB	28762 B	0 KB	00000000
19	a3ec11a72ad38e09	28762 B	0 KB	28850 B	0 KB	00000000

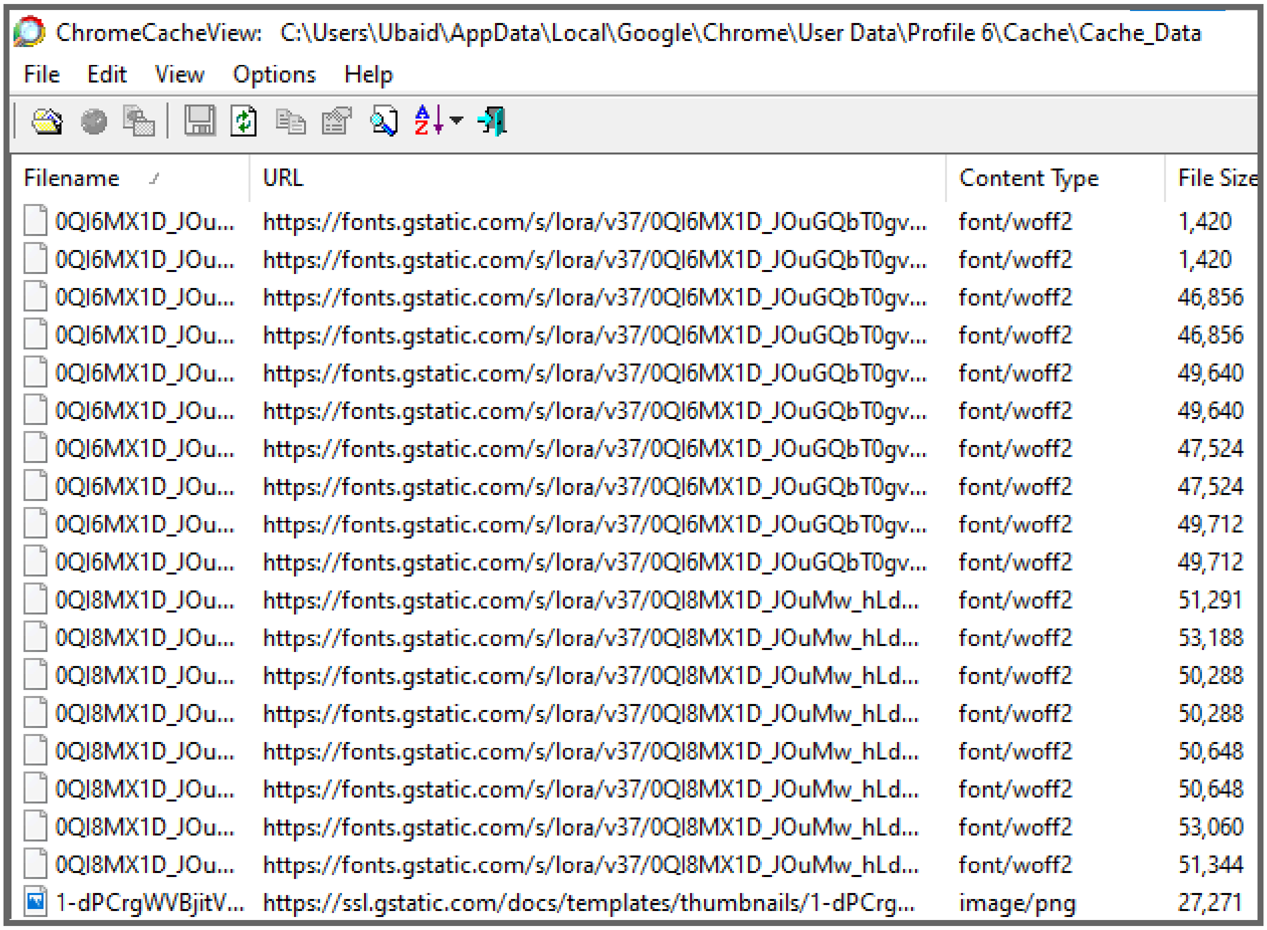
- **Forensic Significance:** Even if the original image is deleted, the thumbcache database proves the user account had visual access to the file.

---

2. Day 58: Browser Cache Extraction (Google Chrome)

Browser cache stores local copies of web content to speed up loading.

- **Analysis Tool:** ChromeCacheView by NirSoft
- **Target Location:** AppData\Local\Google\Chrome\User Data\Default\Cache\Cache_Data
- **Findings:** Parsing the cache revealed extensive records of web activity.
    - **Access Patterns:** A high density of cache writes was identified around 2026-05-11 18:00+ PKT, correlating with the investigator's active session.
    - **Content Types:** Found multiple entries for image/png and application/font-woff2 from domains like gstatic.com and googleusercontent.com.
- **Forensic Significance:** Cache metadata allows an investigator to reconstruct web history independently of the browser's "History" database.



Filename	URL	Content Type	File Size
0QI6MX1D_JOu...	https://fonts.gstatic.com/s/lora/v37/0QI6MX1D_JOuGQbT0gv...	font/woff2	1,420
0QI6MX1D_JOu...	https://fonts.gstatic.com/s/lora/v37/0QI6MX1D_JOuGQbT0gv...	font/woff2	1,420
0QI6MX1D_JOu...	https://fonts.gstatic.com/s/lora/v37/0QI6MX1D_JOuGQbT0gv...	font/woff2	46,856
0QI6MX1D_JOu...	https://fonts.gstatic.com/s/lora/v37/0QI6MX1D_JOuGQbT0gv...	font/woff2	46,856
0QI6MX1D_JOu...	https://fonts.gstatic.com/s/lora/v37/0QI6MX1D_JOuGQbT0gv...	font/woff2	49,640
0QI6MX1D_JOu...	https://fonts.gstatic.com/s/lora/v37/0QI6MX1D_JOuGQbT0gv...	font/woff2	49,640
0QI6MX1D_JOu...	https://fonts.gstatic.com/s/lora/v37/0QI6MX1D_JOuGQbT0gv...	font/woff2	47,524
0QI6MX1D_JOu...	https://fonts.gstatic.com/s/lora/v37/0QI6MX1D_JOuGQbT0gv...	font/woff2	47,524
0QI6MX1D_JOu...	https://fonts.gstatic.com/s/lora/v37/0QI6MX1D_JOuGQbT0gv...	font/woff2	49,712
0QI6MX1D_JOu...	https://fonts.gstatic.com/s/lora/v37/0QI6MX1D_JOuGQbT0gv...	font/woff2	49,712
0QI8MX1D_JOu...	https://fonts.gstatic.com/s/lora/v37/0QI8MX1D_JOuMw_hLd...	font/woff2	51,291
0QI8MX1D_JOu...	https://fonts.gstatic.com/s/lora/v37/0QI8MX1D_JOuMw_hLd...	font/woff2	53,188
0QI8MX1D_JOu...	https://fonts.gstatic.com/s/lora/v37/0QI8MX1D_JOuMw_hLd...	font/woff2	50,288
0QI8MX1D_JOu...	https://fonts.gstatic.com/s/lora/v37/0QI8MX1D_JOuMw_hLd...	font/woff2	50,288
0QI8MX1D_JOu...	https://fonts.gstatic.com/s/lora/v37/0QI8MX1D_JOuMw_hLd...	font/woff2	50,648
0QI8MX1D_JOu...	https://fonts.gstatic.com/s/lora/v37/0QI8MX1D_JOuMw_hLd...	font/woff2	50,648
0QI8MX1D_JOu...	https://fonts.gstatic.com/s/lora/v37/0QI8MX1D_JOuMw_hLd...	font/woff2	53,060
0QI8MX1D_JOu...	https://fonts.gstatic.com/s/lora/v37/0QI8MX1D_JOuMw_hLd...	font/woff2	51,344
1-dPCrgWVBjitV...	https://ssl.gstatic.com/docs/templates/thumbnails/1-dPCrg...	image/png	27,271

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3. Day 59: Recycle Bin Metadata Analysis

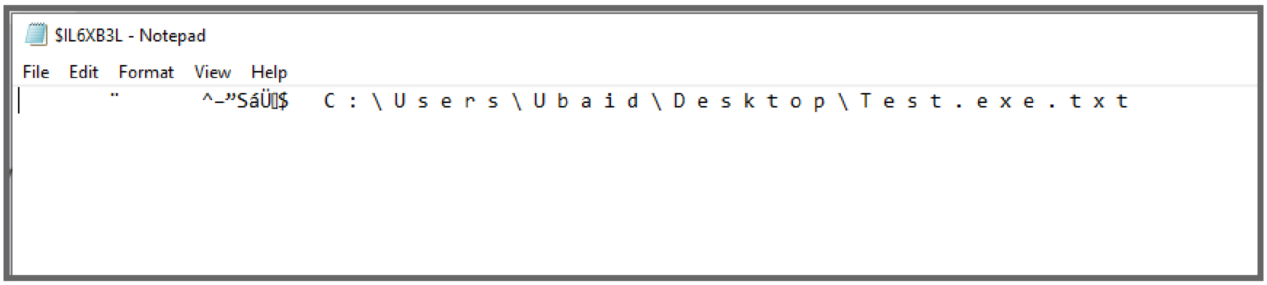
The Recycle Bin retains metadata ($I files) and actual data ($R files) for deleted items.

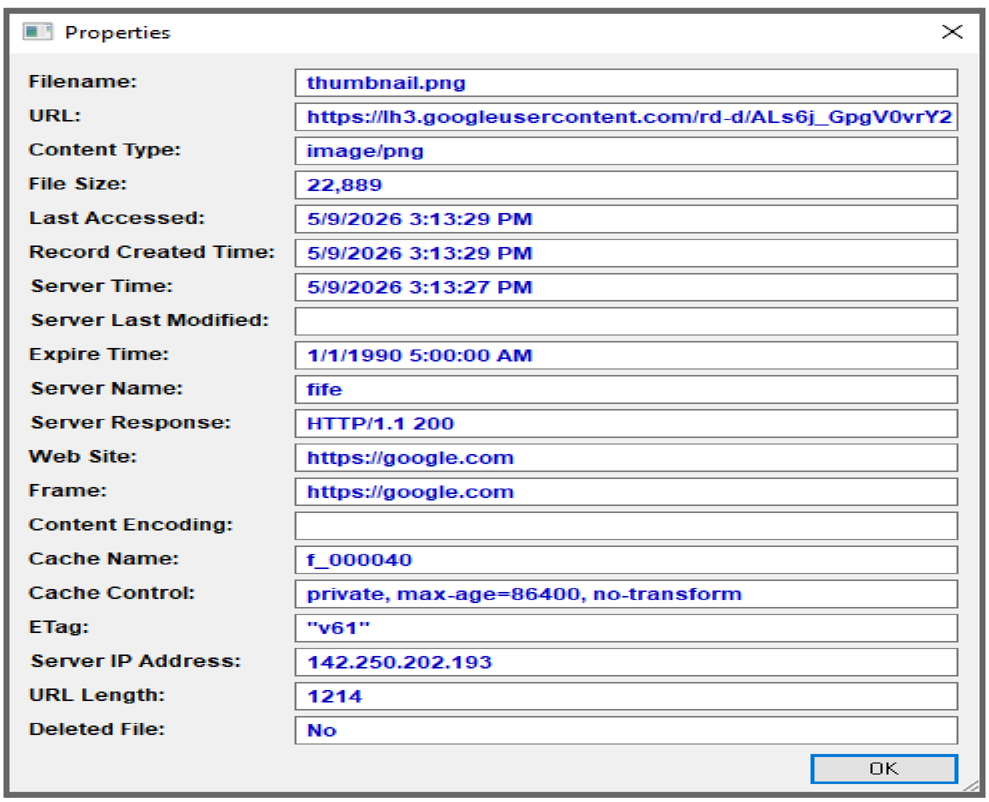
- **Target Directory:** C:\$Recycle.Bin\S-1-5-21-1265788478-1241161198-1512225999-1001
- **Methodology:** The directory was accessed via an elevated terminal using dir /a. The $I metadata file was manually read using Notepad to extract the original file path.

Recycle Bin Log

File Identifier	Original File Path	Deletion Timestamp (PKT)	Data Status ($R)
L6XB3L.txt	C:\Users\Ubaid\Desktop\Test.exe.txt	2026-05-11 07:36 PM	Present & Readable

Analysis: The presence of the $R file confirms the data is recoverable. The $I file definitively links the deleted file to the user's Desktop at the time of deletion.





4. Cross-Artifact Correlation

By correlating all three parts of this week's investigation, we can build a coherent narrative:

1. **Event Logs (Part A):** Verified a successful interactive logon (ID 4624) at 00:39, confirming the user was physically present.
2. **Prefetch (Part B):** Proved that forensic tools (EVTXECMD.EXE) and user applications (CODE.EXE) were executed shortly after logon.
3. **Cache & Recycle Bin (Part C):** Documented the visual assets viewed during the session and the subsequent deletion of files (e.g., Test.exe.txt) from the Desktop.

This synthesized view prevents "isolated arguments" and provides a defensible account of system activity.

End of Week 8 Investigation Report.
